

Research On Triphala Powder

1. Triphala and its constituents ameliorate visceral adiposity from a high-fat diet in mice with diet-induced obesity

Abstract

CONTEXT:

In India, vaidyas (Ayurvedic physicians) traditionally administer triphala and its constituents as therapeutic agents for promoting digestion and satiety.

OBJECTIVE:

The research team performed the present study to investigate the effects of triphala and its constituents (T bellirica [bibhitaki], T chebula [haritaki], and E officinalis [amalaki]) on the dietary induction of obesity (diet-induced obesity [DIO]), and other symptoms of visceral obesity syndrome, in mice fed a high-fat diet (HFD).

DESIGN:

The research team obtained 42 fertile, male, Swiss albino mice, weighing 20 g each, and housed them individually in an approved small-animal facility, in a pathogen-free environment. The team generated DIO mice by feeding them a HFD.

SETTING:

The study took place at the Birla Institute of Technology and Science (BITS) in Pilani, India.

INTERVENTION:

The research team fed all mice, except those in a control group (ND), a HFD for 10 weeks beginning at 7 weeks of age, supplementing the HFDs with herbal treatments for 4 of the groups. The team divided the mice into six weight-matched groups of seven mice each: (1) normal diet (ND), (2) high-fat diet (HFD), (3) triphala (HFD+T), (4) amalaki (HFD+A), (5) haritaki (HFD+H), and (6) bibhitaki (HFD+B).

OUTCOME MEASURES:

The research team evaluated daily energy intake, fasting plasma glucose, serum lipid profile, and liver cytology. The team measured food and energy intake daily for 10 weeks and measured the body weight of each mouse every third day during the course of the experiment. The team drew blood samples at 2, 4, 8, and 10 weeks posttreatment and determined fasting plasma-glucose concentrations and fasting plasma concentrations of cholesterol, triglycerides (TG), LDL, HDL, and plasma alanine transaminase (ALT) using commercial kits. At the completion of the study, a pathologist examined the livers and diagnosed a fatty liver based on the presence of macrovesicular or microvesicular fat in the hepatocytes.

RESULTS:

The research team's results showed that mice fed a HFD for a 10-week period, supplemented with herbal preparation(s) of triphala or its constituents, resulted in significant reductions in body weight ($P < .0001$), energy intake, and percentage of body fat ($P < .001$), as compared with mice in the HFD group. Herbal treatment significantly improved the lipid profiles of the mice by lowering serum total cholesterol (Total-C), TG, and low-density lipoprotein cholesterol (LDL-C) and increasing levels of high-density lipoprotein cholesterol (HDL-C) as compared to the mice in the HFD group. The research team also found that herbal treatment attenuated glucose levels, oral glucose tolerance as measured by the oral glucose tolerance test (OGTT), and levels of ALT. In addition to treatment with its three individual components, treatment

with a popular Ayurvedic formulation of triphala also reversed the pathological changes in liver tissue and decreased the relative weight of visceral adipose fat pads.

CONCLUSIONS:

The present findings suggest that triphala and its constituents can counter the effects of an environment (ie, high dietary intake of fats) and have the potential for use as antiobesity agents with desirable lipid-profile modulating properties.

2. Triphala, Ayurvedic formulation for treating and preventing cancer: a review.

Abstract

BACKGROUND:

Triphala (Sanskrit tri = three and phala = fruits), composed of the three medicinal fruits *Phyllanthus emblica* L. or *Emblica officinalis* Gaertn., *Terminalia chebula* Retz., and *Terminalia belerica* Retz. is an important herbal preparation in the traditional Indian system of medicine, Ayurveda. Triphala is an antioxidant-rich herbal formulation and possesses diverse beneficial properties. It is a widely prescribed Ayurvedic drug and is used as a colon cleanser, digestive, diuretic, and laxative. Cancer is a major cause of death, and globally studies are being conducted to prevent cancer or to develop effective nontoxic therapeutic agents. Experimental studies in the past decade have shown that Triphala is useful in the prevention of cancer and that it also possesses antineoplastic, radioprotective and chemoprotective effects.

CONCLUSIONS:

This review for the first time summarizes these results, with emphasis on published observations. Furthermore, the possible mechanisms responsible for the beneficial effects and lacunas in the existing knowledge that need to be bridged are also discussed.

3. Comparison of enteroprotective efficacy of triphala formulations (Indian Herbal Drug) on methotrexate-induced small intestinal damage in rats.

Triphala is categorized as a rejuvenator and antioxidant-rich Ayurvedic herbal formulation and has traditionally been used in various gastric problems including intestinal inflammation. The aim of the present study was to examine the comparative enteroprotective effect of Triphala formulations against methotrexate-induced intestinal damage in rats. Triphala formulations were prepared by mixing equal (1:1:1) and unequal (1:2:4) proportions of *Terminalia chebula* Retz., *Terminalia belerica* (Gaertn.) Roxb. and *Emblica officinalis* Gaertn. Intestinal damage was induced by administering methotrexate (MTX) in a dose of 12 mg/kg, orally for 4 days to albino rats. The intestinal damage response was assessed by gross and microscopical injury, measuring the intestinal permeability to phenol red and tissue biochemical parameters. Triphala equal and unequal formulations at the dose of 540 mg/kg significantly restored the depleted protein level in brush border membrane of intestine, phospholipid and glutathione content and decreased the myeloperoxidase and xanthine oxidase level in intestinal mucosa of methotrexate-treated rats. In addition, Triphala unequal formulation showed significant decrease in permeation clearance of phenol red with significant attenuation in the histopathological changes, level of disaccharidase in brush border membrane vesicles and lipid peroxidation content of intestinal mucosa. Based on the data generated, it is suggested that Triphala unequal formulation provides significantly more protection than Triphala equal formulation against methotrexate-induced damage in rat intestine.

4.Hypolipidemic effect of triphala in experimentally induced hypercholesteremic rats.

Hypercholesteremia is one of the risk factors for coronary artery disease. The present study highlights the efficacy of Ayurvedic herbal formulation Triphala (*Terminalia chebula*, *Terminalia belerica*, and *Emblca officinalis*) on total cholesterol, Low density lipoprotein (LDL), Very low density lipoprotein (VLDL), High density lipoprotein (HDL) and free fatty acid in experimentally induced hypercholesteremic rats. Four groups of rats were employed namely control, Triphala treated, hypercholesterolemia rats (4% Cholesterol + 1% cholic acid + egg yolk) and Triphala pre-treatment in hypercholesteremic rats. Results showed significant increase in the total cholesterol, LDL, VLDL, and free fatty acid in hypercholesteremic rats were significantly reduced in Triphala treated hypercholesteremic rats. The data demonstrated that Triphala formulation was associated with hypolipidemic effects on the experimentally induced hypercholesteremic rats.

5.Role of Triphala in dentistry

INTRODUCTION

Ayurveda is considered as the “science of life.” The ancient Indian system of health care focused on views of man and his illness.[1] India has an ancient heritage of traditional herbal medicine. Long before the advent of modern medicine, herbs were the mainstream remedies for nearly all ailments. Herbal medicines are being used increasingly as dietary supplements to fight or prevent common diseases.[2] Herbal medicines were in great demand in the developed as well as in developing countries for primary health care because of their wide biological and medicinal activities, higher safety margin, and lower costs.[3] The World Health Organization estimates that about 80% of the populations living in the developing countries rely almost exclusively on traditional medicine for their primary health care needs. Conventional drugs usually provide effective antibiotic therapy for bacterial infections, but there is an increasing problem of antibiotic resistance and a continuing need for new solutions. Hence, now a days, herbal drugs are preferred to synthetic antibiotics.[2]

‘Triphala’ is a well-known powdered preparation in the Indian system of medicine (ISM), being used in Ayurveda since ancient time. Triphala consists of equal parts of the *Emblca officinalis*, *Terminalia chebula*, and *Terminalia belerica*.[4]

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INGREDIENT-WISE MAIN CHEMICAL CONSTITUENTS OF TRIPHALA

Tannins

“Tannin” is a general descriptive name for a group of polymeric phenolic substances capable of tanning leather or precipitating gelatin from solution, a property known as astringency. This

group of compounds, especially green teas and red wines, has received a great deal of attention in recent years since they can cure or prevent a variety of ills. Many human physiological activities, such as stimulation of phagocytic cells, host-mediated tumor activity, and a wide range of anti-infective actions, have been assigned to tannins. One of their molecular actions is to complex with proteins through so-called non-specific forces such as hydrogen-bonding and hydrophobic effects, as well as by covalent bond formation. Thus, their mode of anti-microbial action may be related to their ability to inactivate microbial adhesins, enzymes, and cell envelope transport proteins.[5]

Quinones

Quinones are aromatic rings with two ketone substitutions. They are ubiquitous in nature and are characteristically highly reactive. The individual redox potential of the particular quinone-hydroquinone pair is very important in many biological systems. Vitamin K is a complex naphthoquinone with anti-hemorrhagic activity. In addition to providing a source of stable free radicals, quinones are known to complex irreversibly with nucleophilic amino acids in proteins, often leading to inactivation of the protein and loss of function. For that reason, the potential range of quinone anti-microbial effects is great. Probable targets in the microbial cell are surface-exposed adhesins, cell wall polypeptides, and membrane-bound enzymes. Quinones may also render substrates unavailable to the microorganism.[5]

Flavones, flavonoids, and flavonols

Flavones are phenolic structures containing one carbonyl group (as opposed to the two carbonyls in quinones). The addition of a 3-hydroxyl group yields a flavonol. Flavonoids are also hydroxylated phenolic substances, but occur as a C6-C3 unit linked to an aromatic ring. Since they are known to be synthesized by plants in response to microbial infection, it should not be surprising that they have been found *in vitro* to be effective anti-microbial substances against a wide array of microorganisms. Their activity is probably due to their ability to complex with extracellular and soluble proteins and to complex with bacterial cell walls. More lipophilic flavonoids may also disrupt microbial membranes. These compounds have been shown to inhibit *Vibrio cholera* O1, *Shigella*, *Streptococcus mutans* *in vitro*. Inhibition of isolated bacterial glucosyltransferases in *S. mutans*, and reduction of fissure caries by about 40% has also been demonstrated.[5]

Gallic acid

Gallic acid is a common phyto-constituent present in all three herbs used in Triphala. It is reported to possess hepatoprotective and antioxidant activity. It also suppresses growth of cancer cells.[4]

Vitamin C

Fruit juice of *Emblica officinalis* (EO) contains the highest vitamin C (478.56 mg/100 mL) content. The fruit when blended with other fruits boosted their nutritional quality in terms of vitamin C content. Vitamin C in EO accounts for approximately 45-70% of the antioxidant activity.[6] Evidences have been reported for the relation between vitamin C and periodontal disease. Significant gum bleeding can occur in vitamin C deficiency. Vitamin C along with bioflavonoid helps to speed up the healing process.[7]

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INDIVIDUAL COMPONENTS OF TRIPHALA

Emblica officinalis (Amalaki)

(Individual chemical ingredient: Vitamin C, carotene, nicotinic acid, riboflavin, and tannins).[8]

Amalaki is known by the botanical name *Emblica officinalis* and also known in Sanskrit as Dhatri (The nurse), which is a reference to its incredible healing properties. Amalaki can be taken individually in powder form, a decoction or as a confection. Amalaki fruit is known to be one of the best rasayanas in Ayurveda, with anti-oxidant and anti-aging properties. It has its beneficial role in cancer, diabetes, liver treatment, heart trouble, ulcer, anemia, and various other diseases. Similarly, it has application as immunomodulatory, anti-pyretic, analgesic, cytoprotective, anti-tussive, and gastroprotective agent. Additionally, it is useful in memory enhancing, ophthalmic disorders, and lowering cholesterol level. It is also helpful in neutralizing snake venom and as an anti-microbial agent against *Escherichia coli*, *K. ozaenae*, *Klebsiella pneumoniae*, *Proteus mirabilis*, *Pseudomonas aeruginosa*, *S. paratyphi A*, *S. paratyphi B* and *Serratiamarcescens*. The drug is not reported to have any side-effects even after prolonged use.[6]

Terminalia chebula (Hiritaki or Black myrobalan)

(Individual chemical ingredient: Tannins, anthraquinones, and polyphenolic compounds).[8]

Terminalia chebula is a plant species belonging to the genus *Terminalia*, family Combretaceae. The fruit of the tree has been used as traditional medicine for household remedy against various human ailments, since antiquity. *Terminalia chebula* has been extensively used in Ayurveda, Unani, and Homoeopathic medicine and has become a cynosure of modern medicine. *Terminalia chebula* is rich in tannin. The chief constituents of tannin are chebulic acid, chebulagic acid, corilagin, and gallic acid.

Terminalia chebula exhibited anti-bacterial activity against a number of Gram-positive and Gram-negative human pathogenic bacterial species. It also exhibits anti-fungal and anti-viral

properties. It has also shown anti-mutagenic/anti-carcinogenic activity, antioxidant activity, adaptogenic and anti-anaphylactic activities, immunomodulatory activity, cytoprotective and radioprotective activity. It is also effective in hypolipidemia/hypercholesterolemia, improving gastro-intestinal motility with anti-spasmodic activity, diabetes, retinopathy, and wound healing.[3]

***Terminalia bellerica* (Bibhitaki)**

(Individual chemical ingredient: Gallic acid, tannic acid, and glycosides).[8]

Terminalia bellerica Roxb. (Combretaceae), commonly known as “belleric myrobalan” and locally as “bahera,” is a large deciduous tree, found throughout central Asia and some other parts of the world. Its fruit is used in folk medicine to treat asthma, cancer, colic, diarrhea, dysuria, headache, hypertension, inflammations, and pain. The plant is reported to contain termilignan, thannilignan, anolignan B, gallic acid, ellagic acid, β -sitosterol, arjungenin, belleric acid, bellericosidem, flavonoids, and tannins. *T. bellerica* possesses antioxidant, anti-spasmodic, bronchodilatory, hypercholesterolemic, anti-bacterial, cardioprotective, hepatoprotective, hypoglycemic, and hypotensive properties.[9]

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MARKETED FORMULATION OF TRIPHALA

Triphala tablet, Triphala choorna

Formulation of choornam: This is a dry fine powder form of the drug choornam, which can be used both internally and externally.

Preparation: The drug selected is washed cleaned and dried. It is crushed to a fine powder using a pulverizer. The fineness of the powder improves the therapeutic efficacy. In case of compound choorna, each drug should be powdered separately, and finally all individual drug powders are mixed together. The choorna should be fine of at least 80 mesh sieve.[10]

Decoction form: This form can be used as an eyewash or mouthwash.

Preparation: After cleaning the dried fruits and removing seeds, the powder is made separately from the three dried fruits. Three powders are mixed together in equal amounts to form a uniform mixture. This mixture is added to 16 times water for an hour and then boiled till half of the water remains. The decoction is filtered through fine cotton cloth and stored in a clean bowl or jug. Slightly warm decoction should be used for washing eyes at the earliest after its preparation.[10]

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TRIPHALA IN DENTISTRY

Anti-caries activity

Despite several anti-plaque agents available in the market, the search for an effective agent still continues. Several undesirable side-effects associated with these agents stimulated the search for alternate agents. Plants or plant products used in folk dental practices or prescribed in Unani, homeopathic, or Ayurvedic remedies are now gaining attention in view of their acclaimed medicinal properties.

Terminalia chebula is valuable in the prevention and treatment of several diseases of the mouth such as dental caries, spongy and bleeding gums, gingivitis, and stomatitis. The extract could successfully prevent plaque formation on the surface of the tooth, as it inhibited the sucrose-induced adherence and the glucan-induced aggregation, the two processes which foster the colonization of the organism on the surface of the tooth. Thus, the extract of *T. chebula* may be an effective agent in the treatment of carious teeth, owing to its ability to inhibit the growth and accumulation of *S. mutans* on the surface of the tooth. This would prevent the accumulation of acids on the surface of the tooth, and thus the further demineralization and the breakdown of the tooth enamel.[\[11\]](#)

Triphala as a root canal irrigant

Primary endodontic infections are caused by oral microorganisms, which are usually opportunistic pathogens that may invade a root canal containing necrotic tissue and establish an infectious process. The number of facultative anaerobic bacteria increases when the root canal remains infected for long periods. *Enterococcus faecalis*, a facultative anaerobic gram-positive coccus, is the most common *Enterococcus* sp. cultured from non-healing endodontic cases. Sodium hypochlorite (NaOCl) is an efficient irrigant used in eliminating *E. faecalis* biofilms *in vitro*, but its main disadvantages are its unpleasant taste, high toxicity, and its inability to remove the smear layer. Triphala has shown significant anti-bacterial activity against three and six week biofilms. The use of herbal alternatives as a root canal irrigant might prove to be advantageous considering the several undesirable characteristics of NaOCl.[\[12\]](#)

Anti-collagenase activity of Triphala

Matrix metalloproteinases play a vital role in periodontal destruction, and this knowledge lead to a new concept involving the chemotherapeutic inhibition of these enzymes. Doxycycline is most potent tetracycline for collagenase/gelatinase inhibition. However, long-term tetracycline therapy has certain disadvantages. Use of herbal product extract in treating periodontal disease does not produce side-effects of tetracycline compounds as well as other synthetic drugs. Triphala has

strong inhibitory activity against PMN-type collagenase, especially MMP-9 at a 1500 µg/ml concentration, which is well within the safety profile of toxicological studies.[13]

Anti-microbial and anti-oxidant effect of Triphala

Anti-microbial and anti-oxidant effect of Triphala has been proven *in-vitro* as it has been shown to inhibit *Streptococcus mutans* at concentrations as low as 50µg/ml. This anti-plaque effect probably may be due to the tannic acid in Triphala, which is adsorbed well to the groups on the surface of the bacterial cells, which result in protein denaturation and ultimately to bacterial cell death.

The strong antioxidant activity of Triphala may be attributed to *T. belerica*, which is the most active antioxidant followed by *E. officinalis* and *T. chebula*. The major ingredients of *T. belerica* are ellagic and gallic-acid; *E. officinalis* has several gallic acid derivatives including epigallocatechingallate and in *T. chebula*, gallic acid is the major ingredient. The presence of these active ingredients of phenolic nature may be responsible to scavenge the free radicals.[14]

Triphala as a mouth rinse

Ayurvedic drugs have been used since ancient times. Oral rinses made from these are used in periodontal therapy. Triphala is one of these with wide spectrum of activity. According to the Sushruta Samhita, Triphala can be used as a gargling agent in dental diseases.

0.6% Triphala mouthwash has shown to have significant anti-caries activity, which is comparable to that of chlorhexidine without possessing disadvantages as staining of teeth and at much less cost although there was no evidence of re-mineralization of tooth structure.[15]

Triphala mouth rinse when combined with scaling and root planing showed significant reduction in the plaque, gingival, and oral hygiene indices without any evidence of staining of teeth at seven, 30, and 45 days, which was comparable to reduction obtained by chlorhexidine mouth rinse in combination with scaling and root planing.[7]

Triphala mouthwash twice-daily combined with metronidazole 400 mg thrice-daily when compared with 0.2% chlorhexidine with metronidazole 400 mg thrice-daily and Triphala mouthwash with oral powder of Triphala in a one month study showed improvement in clinical indices in terms of reduction in tooth mobility, pocket depth, bleeding gums, sensitivity to hot and cold, and calculus formation with minimal recurrence in all the clinical parameters.[16]

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.CONCLUSION

Triphala is a novel drug with an array of therapeutic activities gifted by Ayurveda to the world. It has potential to treat a variety of human ills with minimal or no side-effects. Dentistry is still in search of a drug for diseases affecting hard and soft tissues of oral cavity. Triphala seems to fulfill most of these requirements without any adverse effect on oral tissues and at very minimal cost as compared to commercially available products today. Hence, further research exploring various therapeutic actions of Triphala should be encouraged in dentistry.

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Footnotes

Source of Support: Nil

Conflict of Interest: None declared.

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REFERENCES

1. Samy RP, Pushparaj PN, Ponnampalam G. A compilation of bioactive compounds from Ayurveda. *Bioinformation*. 2008;3:100–10. [[PMC free article](#)] [[PubMed](#)]
2. Tambekar DH, Dahikar SB, Lahare MD. Antibacterial potentials of some herbal preparations available in India. *Res J Med Med Sci*. 2009;4:224–7.
3. Chattopadhyay RR, Bhattacharyya SK. *Terminalia chebula*: An update. *Pharmacogn Rev*. 2007;1:151–6.
4. Mukherjee PK, Rai S, Bhattacharyya S, Debnath P, Biswas TK, Jana U, et al. Clinical Study of ‘Triphala’: A well known phytomedicine from India. *Iran J Pharmacol Ther*. 2006;5:51–4.
5. Cowan MM. Plant products as antimicrobial agents. *Clinical Microbiology Reviews*. 1999;12:564–82. [[PMC free article](#)] [[PubMed](#)]
6. Khan KH. Roles of *Embolia officinalis* in medicine: A review. *Bot Res Int*. 2009;2:218–28.
7. Desai A, Anil M, Debnath S. A clinical trial to evaluate the effects of Triphala as a mouthwash in comparison with chlorhexidine in chronic generalized periodontitis patient. *Indian J Dent Adv*. 2010;2:243–7.
8. India: World Health Organization; 2010. WHO, 2010. Traditional Herbal Remedies for Primary Health Care.

9. Khan A, Gilani AH. Antisecretory and analgesic activities of *Terminalia bellerica*. Afr J Biotechnol.2010;9:717–9.
10. Bele AA, Jadhav VM, Kadam VJ. Potential of Tannins: A Review. Asian J Plant Sci. 2010;9:209–14.
11. Jagtap AG, Karkera SG. Potential of the aqueous extract of *Terminalia chebula* as an anticaries agent. J Ethnopharmacol. 1999;68:299–306. [[PubMed](#)]
12. Prabhaka, Senthilkumar M, Priya MS, Mahalakshmi K, Sehgal PK, Sukumaran VG. Evaluation of antimicrobial efficacy of herbal alternatives (Triphala and Green Tea polyphenols), MTAD, and 5% Sodium Hypochlorite against *Enterococcus faecalis* Biofilm Formed on Tooth Substrate: An *In Vitro* Study. J Endod.2010;36:83–6. [[PubMed](#)]
13. Abraham S, Kumar MS, Sehgal PK, Nitish S, Jayakumar ND. Evaluation of the inhibitory effect of Triphala on PMN-type Matrix Metalloproteinase (MMP-9) J Periodontol. 2005;76:497–502. [[PubMed](#)]
14. Jagdish L, Anand Kumar VK, Kaviyarasan V. Effect of Triphala on dental biofilm. Indian J Sci Technol.2009;2:30–3.
15. Tandon S, Gupta K, Rao S, Malagi KJ. Effect of Triphala mouthwash on caries status. Int J Ayurveda Res. 2010;1:93–9. [[PMC free article](#)] [[PubMed](#)]
16. Maurya DK, Mittal N, Sharma KR, Nath G. Role of triphala in the management of Periodontal disease. Anc Sci Life. 1997;17:120–7. [[PMC free article](#)] [[PubMed](#)]

6. Immunomodulatory activity of triphala on neutrophil functions.

Abstract

Immune activation is an effective as well as protective approach against emerging infectious diseases. The immunomodulatory activities of Triphala (*Terminalia chebula*, *Terminalia bellerica* and *Embellica officinalis*) were assessed by testing the various neutrophil functions like adherence, phagocytosis (phagocytic index (P.I) and avidity index (A.I)) and nitro blue tetrazolium (NBT) reduction in albino rats. In recent years much attention is being focused on the immunological changes occur during stress. Noise (100 dB) stress for 4 h/d for 15 d, was employed to alter the neutrophil functions. The neutrophil function tests and corticosterone levels were carried out in eight different groups of animals, namely control, Triphala, noise-stress, Triphala noise-stress, and corresponding immunized groups were used. Sheep red blood cells (SRBC 5 x 10⁹ cells per ml) were used for immunizing the animals that belongs to immunized groups. In Triphala administration (1 g/kg/d for 48 d), A.I was found to be significantly enhanced in the Triphala group, while the remaining neutrophil functions and steroid levels were not altered significantly. However the neutrophil functions were significantly enhanced in the Triphala immunized group with a significant decrease in corticosterone level was observed. Upon exposure to the noise-stress, the neutrophil functions were significantly suppressed and followed by a significant increase in the corticosterone levels were observed in both the noise-stress and the noise-stress immunized groups. These noise-stress-induced changes were significantly prevented by Triphala administration in both the Triphala noise-stress and the Triphala noise-stress immunized groups. Hence our study has divulged that oral administration of Triphala appears to stimulate the neutrophil functions in the

immunized rats and stress induced suppression in the neutrophil functions were significantly prevented by Triphala.

7. Comparison of Gastroprotective Effects of *Triphala* Formulations on Stress-induced Ulcer in Rats

Abstract

Peptic ulcer is the most common gastrointestinal disorder in clinical practice. There is evidence concerning the participation of reactive oxygen species in the etiology and pathophysiology of human diseases, such as neurodegenerative disorders, inflammation, viral infections, autoimmune pathologies and digestive system disorders such as gastrointestinal inflammation and gastric ulcer[1]. Study have shown that alteration in the antioxidant status following stress-induced ulcer indicate free radicals are directly implicated in the mechanism of stress-induced ulceration in rats[2]. Considering the several side effects (arrhythmias, impotence, gynaecomastia and hematopoietic changes) of modern medicine, indigenous drugs possessing fewer side effects should be looked for as a better alternative for the treatment of peptic ulcer[3].

The edifice of the drug science of Ayurveda stands on a strong foundation of the basic fundamentals of *pancha-mahabhutas* and *tridosha*. The three *doshas*, namely *vata*, *pitta* and *kapha* are biological representatives for physiological functions in the state of homeostasis and for pathological disorders in the state of imbalance[4]. The vitiation of *pitta dosha* lead to impairment of *agni* resulted in to *amlapitta*(hyperacidity), *grahani roga* (malabsorption syndrome) and other gastrointestinal disorders. Triphala formulation is one of the renowned Ayurvedic formulation used alone or along with other ingredients in Ayurvedic therapeutics for the treatment of gastrointestinal problems. It is categorized as tridoshik rejuvenator and reported to be an antioxidant rich herbal formulation[5,6]. As per Ancient text, one of the *Triphala* formulations called as *Chinnodbhavadi kwath* (decoction) is used for chronic hyperacidity and gastric problems[7]. Pharmacological studies have shown that *Triphala* extract possess antioxidant activity and reduce the damage due to oxidative stress[8]. It has been reported to be cytotoxic to breast cancer cells and prostate cancer cells[9], radio protective[6] and displays antidiabetic and free radicals scavenging activities[10].

The present study was thus aimed to investigate the comparative gastroprotective effects of *Triphala* formulations in stress-induced gastric ulcer in rats to determine which of the two formulation- *Triphala* equal or *Triphala* unequal is better for the above property and to ascertain whether this property is retained when *Triphala* formulation used as an ingredient of *Chinnodbhavadi kwath* (decoction) to substantiate its traditional claim.

Fruits of *Terminalia chebula* Retz. (Combretaceae), *Terminalia belerica* (Gaertn) Roxb. (Combretaceae) and *Emblica officinalis* Gaertn. (Euphorbiaceae) were collected from forest of

Dang and Valsad (Gujarat, India) in the month of December (2005). Stem of *Tinospora cordifolia* (Willd.) Miers. (Menispermaceae), stem bark of *Azadirachta indica* A. Juss. (Meliaceae) and leaves of *Trichosanthes dioica* Roxb. (Cucurbitaceae) were collected from forest of Barda hills, Jamnagar (Gujarat, India) in the month of September and October 2005. The plant materials were authenticated and voucher specimens of each submitted to phamacognosy laboratory of Institute of Postgraduate Teaching and Research, Gujarat Ayurved University, Jamnagar, India. *Triphala* unequal formulation was prepared by mixing one part of *T. chebula*, two parts of *T. belerica* and four parts of *E. officinalis*[5,11] and *Triphala* equal formulation was prepared by mixing these three ingredients in equal proportion (1:1:1)[12]. *Chinnodbhavadi kwath* (decoction) was prepared by mixing equal proportion of *T. chebula*, *T. belerica*, *E. officinalis*, *T. cordifolia*, *A. indica* and *T. dioica*. Coarse powder (48 g) of mixture and 768 g water was added; boiled on low to medium heat till the liquid portion was reduced to 1/8th of the original volume (96 g) and filtered[11]. All chemicals used in the study and for biochemical assay were of analytical grade.

Triphala formulations and their ingredients were standardized using gallic acid as a marker compound by HPTLC finger print. The plate was developed in toluene:ethyl acetate:formic acid (5:5:1) solvent system. Gallic acid was observed at 0.52 R_f value, when scanned at 254 nm. *E. officinalis*, *T. belerica*, *T. chebula* and all three *Triphala* formulations shows almost the same R_f values as observed for gallic acid (fig. 1). The concentration of trace heavy metals such as lead, cadmium, arsenic and mercury present in formulations were analyzed by Atomic Absorption Spectrophotometer. The data obtained indicated that trace metals do not seem to be present in significant quantities in *Triphala* equal, *Triphala* unequal formulations and *Chinnodbhavadi kwath*.

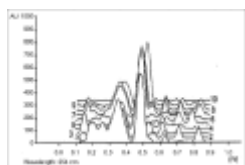


Fig. 1

Comparative HPTLC fingerprints of *Triphala* formulations and their ingredients with gallic acid. 1- *E. officinalis*, 2- *T. belerica*, 3- *T. chebula*, 4- *A. indica*, 5- *T. cordifolia*, 6- *T. dioica*, 7- *Triphala* equal formulation, 8- *Triphala* unequal formulation, ...

Adult Wistar rats (180 to 220 g) of either sex were used in the experiment. The animals were maintained under standard conditions of temperature, humidity and exposed to 12 h light and dark cycles. All animals were exposed to the same environmental conditions and were maintained on standard diet and water *ad libitum*. The experimental protocol was approved by

the Institutional Animal Ethical Committee as per guidelines of Committee for the Purpose of Control and Supervision on Experiments on Animals, India.

Acute oral toxicity of *Triphala* formulations were carried out in female rats as per the 423 guideline of OECD (Organization of Economic Co-operation and Development)[13]. The formulations were tested at limit dose of 2000 mg/kg body weight of female rats. The result showed that *Triphala* formulations did not produce any changes in observed parameters and there was no mortality at the dose level tested. Hence, animal dose of *Triphala* formulations were fixed on the basis of human therapeutic dose mentioned in literatures[11,12].

Rats were randomized in to six groups, each consisting of six animals. Group (I) normal control group and Group (II) stress control group, received vehicle as an aqueous suspension of 1% carboxymethyl cellulose (CMC) in the dose of 10 ml/kg body weight. Rats of group (III) and (IV) received *Triphala* equal and *Triphala* unequal formulations, respectively in a dose of 540 mg/kg. *Triphala* formulations were suspended in 1% CMC and administered orally once daily for seven consecutive days to respective groups. Rats of group (V) were treated with *Chinnodbhavadi kwath* in a dose of 8.7 ml/kg for seven days. Group (VI) was treated with omeprazole 48, 24 and 1 h prior to induction of ulcers in a dose of 20 mg/kg and used as a positive control group.

Water immersion stress-induced gastric ulcer was induced by following the method described earlier[14]. On seventh day, the individually over night fasted rats were exposed to the water immersion stress inside specially arranged containers, which were made up of plexi-glass with holed lids. The rats were sacrificed at the end of 14 h period after water immersion stress. Abdominal cavity was opened carefully and the stomach was excised for assessment of ulcer index and tissue biochemical studies.

The stomach was excised, cleaned and opened along its greater curvature and the inner surface gently washed with cold saline solution and examined for ulceration with a magnifying lens. Severity and total number of ulcers in each rat were recorded for calculating ulcer index[15]. The scoring was done as, 0- no visible ulcer, 1- maximum diameter of 1 mm, 2- maximum diameter of 1-2 mm, 3- maximum diameter of 2-3 mm, 4- maximum diameter of 3-4 mm, 5- maximum diameter of 4-5 mm, 10- An ulcer over 5 mm in diameter, 25- A perforated ulcer.

The stomach was subsequently divided in to longitudinal pieces and stored immediately at -20° for estimation of tissue biochemical parameters. Protein content was quantitated using bovine serum albumin as a standard and was expressed as mg/g wet tissue[16]. Lipid peroxidation (LPO) was measured as thiobarbituric acid reactive substances (TBARS) formation[17]. TBARS concentrations were calculated by the use of malondialdehyde (MDA) as a standard and results were expressed as nmol MDA/g tissue. Hydroxyl radical ion react with thiobarbituric acid

resulted in to pink colour which was estimated and the results were expressed as Unit/mg protein[18].

Superoxide dismutase (SOD) activity was determined by the nitro blue tetrazolium reduction method[19]. One unit of enzyme activity that inhibit rate of reaction by 50% in one minute under the defined assay conditions and the results have been expressed as Unit/mg protein. Catalase in stomach tissue was measured and content was expressed as $\mu\text{moles H}_2\text{O}_2$ consumed/mg protein/min[20]. Glutathione present was estimated by the method of Grunert and Phillips[21] and the results were expressed as nmol/g wet tissue.

The estimation of adenosine triphosphatases includes $\text{Na}^+\text{-K}^+\text{-ATPase}$ [22], $\text{Ca}^{2+}\text{-ATPase}$ [23] and $\text{Mg}^{2+}\text{-ATPase}$ [24]. The concentrations of ATPase were expressed as μmoles of phosphorus liberated /mg protein/min at 37° . Deoxyribonucleic acid and ribonucleic acid was estimated in stomach mucosa, homogenized in trichloro acetic acid and extracted with alcohol diethyl ether reagent as per the method described previously[25].

The data are expressed as mean $\pm$ standard error of mean for six rats per experimental group. One way analysis of variance (ANOVA) was used to compare the mean values of quantitative variables among the groups followed by Dunnet's multiple *t*-test for unpaired data to determine significant difference between groups at $P<0.05$.

Triphala is considered as one of the most important formulation in Ayurvedic therapeutics for its multiple organ protective effects including gastroprotection. Various physical and psychological stresses cause gastric ulceration in humans and experimental animals. Oxidative damage is considered to be common factor in the pathogenesis of ulcers by different experimental and clinical models. The increases in free radicals generation during stress are held responsible for induction of ulcers[26]. In the present study, rats subjected to water immersion stress showed significant increased in ulcer index as shown in (Table 1). Rats pretreated with *Triphala* unequal formulation ($P<0.05$) and *Chinnodbhavadi kwath* ($P<0.05$) showed significant reduction in ulcer index when compared with the stress control group. *Triphala* equal formulation exhibited a moderate non-significant decrease in ulcer index.

Groups	Ulcer Index	LPD ($\mu\text{mol H}_2\text{O}_2/\text{g}$ tissue)	Total protein (mg/g tissue)	Protein (mg/g tissue)	SOD (U/mg protein)	Catalase (U/mg protein)
Normal control	0	0.00 $\pm$ 0.01	0.10 $\pm$ 0.01	0.10 $\pm$ 0.01	1.00 $\pm$ 0.01	1.00 $\pm$ 0.01
Stress control	17.00 $\pm$ 0.02	0.10 $\pm$ 0.02	0.10 $\pm$ 0.02	0.10 $\pm$ 0.02	1.00 $\pm$ 0.02	1.00 $\pm$ 0.02
Triphala equal	15.00 $\pm$ 0.04	0.10 $\pm$ 0.04	0.10 $\pm$ 0.04	0.10 $\pm$ 0.04	1.00 $\pm$ 0.04	1.00 $\pm$ 0.04
Triphala unequal	14.00 $\pm$ 0.03	0.10 $\pm$ 0.03	0.10 $\pm$ 0.03	0.10 $\pm$ 0.03	1.00 $\pm$ 0.03	1.00 $\pm$ 0.03
Chinnodbhavadi kwath	9.00 $\pm$ 0.01	0.10 $\pm$ 0.01	0.10 $\pm$ 0.01	0.10 $\pm$ 0.01	1.00 $\pm$ 0.01	1.00 $\pm$ 0.01
Omeprazole	4.00 $\pm$ 0.01	0.10 $\pm$ 0.01	0.10 $\pm$ 0.01	0.10 $\pm$ 0.01	1.00 $\pm$ 0.01	1.00 $\pm$ 0.01

The values are expressed as mean $\pm$ S.E. of six rats per group. * $P<0.05$, compared with stress control group. ** $P<0.01$, compared with stress control group.

TABLE 1

EFFECT OF *TRIPHALA* FORMULATIONS ON ULCER INDEX AND GASTRIC MUCOSAL BIOCHEMICAL PARAMETERS IN STRESS-INDUCED ULCER IN RATS

It is observed that stress procedure significantly elevated the lipid peroxidation and hydroxyl radicals ($P<0.05$) and concomitantly reduced the activity of SOD, catalase and total glutathione ($P<0.05$) content in gastric mucosa of rats which is indicative of oxidative stress ([Table 1](#)). Cell membrane lipid is very susceptible to hydroxyl radical attack and initiates the formation of LPO. *Triphala* unequal formulation and *Chinnodbhavadi kwath* treatment improve stomach oxidative balance in rats because they were able to reduce the level of hydroxyl radical and malondialdehyde, a good indicator of lipid peroxidation[17]. *Triphala* unequal formulation and *Chinnodbhavadi kwath* significantly reversed the stress-induced free radical generation. This may be due to restoration of free radicals scavenging enzymes viz; SOD, catalase and glutathione in gastric mucosa [Table 1](#), effectively counter acting the free radicals generated during the stress condition. Earlier study also revealed that biological activities of *Triphala* due to its ability to scavenge free radicals[7]. The presence of the *in vitro* antioxidant activity of *Triphala* by scavenging oxygen radicals together with the inhibition of lipid peroxidation due to presence of phenolic compound particularly gallic acid[9]. Further, the flavanoid isolated from *E. officinalis* is reported to lower the LPO level in rats[27].

In rats subjected to stress, there was significant decrease in the level of membrane bound Na^+ -ATPase, Ca^{2+} -ATPase and Mg^{2+} -ATPase ($P<0.05$) in comparison to normal control group ([Table 2](#)). *Chinnodbhavadi kwath* significantly reversed decreased level of membrane bound ATPase near to normal control group in stress condition. The effect might be due to decrease in LPO and restoration of the level of GSH significantly and thereby protecting the protein thiol and sulfahydryl groups which are essential for structural integrity and function of ATPase. It is pertinent here to refer to a previous report which suggested that *Emblica officinalis* extract[28] and *Azadirachta indica*[29] significantly prevented depletion of nonprotein sulfahydryl groups, that might have contributed to the protective effect of *Chinnodbhavadi kwath*.

Groups	Na ⁺ -ATPase activity (μg liberated/mg protein/min)	Ca ²⁺ -ATPase activity (μg liberated/mg protein/min)	Mg ²⁺ -ATPase activity (μg liberated/mg protein/min)	DNA %
Normal control	1.15±0.08	1.45±0.14	1.45±0.14	444.2
Stress control	2.16±0.08 ^a	1.25±0.10 ^a	1.35±0.11 ^a	228.26
Triphala equal	2.16±0.13	0.95±0.11	1.25±0.07	291.4
Triphala unequal	2.05±0.08	1.25±0.10	1.45±0.13	368.76
Chinnod. kwath	2.16±0.11 ^b	2.15±0.11 ^b	2.25±0.10 ^b	368.76
Chinnod. kwath	2.16±0.11 ^b	2.15±0.11 ^b	2.25±0.10 ^b	368.76

^a Values are significantly different from normal group. ^b Values are compared with stress control group. ^c P<0.05 vs

EFFECT OF TRIPHALA FORMULATIONS ON GASTRIC MUCOSAL BIOCHEMICAL PARAMETERS IN STRESS-INDUCED ULCER IN RATS

[TABLE 2](#)

EFFECT OF *TRIPHALA* FORMULATIONS ON GASTRIC MUCOSAL BIOCHEMICAL PARAMETERS IN STRESS-INDUCED ULCER IN RATS

Increase or decrease in life span of mucosal cells can be expressed as amount of DNA and RNA in the gastric wall mucosa. The increased DNA and RNA content of gastric wall mucosa to some

extent in *Triphala* unequal formulation and *Chinnodbhavadi kwath* treated groups indicate decreased cell shedding and increased life span of cells[30]. A recent study suggested that phenolic compound, ascorbic acid and flavanoids in *Triphala* formulations are responsible for the protection of DNA[6].

Previous report suggested that *E. officinalis*, *T. cordifolia* and *A. indica* produced significant antiulcer activity[31]. *E. officinalis* produced antiulcer activity due to presence of saponin and tannin[28]. The major antiulcer compound isolated from the aqueous extract of *A. indica* characterized to be phenolic glycoside in nature[32]. *Triphala* formulation is rich source of tannins, which is known to affect the integrity of mucus membrane. Tannins with their protein precipitating and vasoconstriction effects could be advantageous in preventing ulcer development[28]. Further, *Triphala* formulations contains *Terminalia belerica* and *Emblica officinalis* which are well established major antioxidants and their free radical scavenging effect is due to the presence of ellagic acid and gallic acid compared to *Terminalia chebula*[10].

From the present study, it is concluded that *Triphala* formulations possess significant anti-ulcer activity. This activity depends mainly on inhibition of free radical generation due to restoration of free radicals scavenging enzymes, enhancing the stability of gastric mucosal barrier and gastric cytoprotection against stress-induced gastric ulceration in rats. Among the three formulations studied *Chinnodbhavadi kwath* and *Triphala* unequal formulation provides significant protection against gastric ulcer as compared to *Triphala* equal formulation. The reason for the observed efficacy of this preparation is the increased quantity of two of its ingredients namely *Terminalia belerica* and *Emblica officinalis* in *Triphala* unequal formulation. Of the two, *Emblica officinalis* is well-known for its cytoprotective effect. *Chinnodbhavadi kwath* produced antiulcer effect due to the presence of *Triphala* formulation along with *T. cordifolia* and *A. indica*.

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Footnotes

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REFERENCES

1. Repetto MG, Llesuy SF. Antioxidant properties of natural compounds used in popular medicine for gastric ulcers. *Braz J Med Biol Res.* 2002;35:523–34. [[PubMed](#)]
2. Sood S, Muthuraman A, Gill NS, Bali M, Sharma PD. Effect of *Citrus karna* peel extract on stress-induced peptic ulcer in rat. *J Biol Sci.* 2010;10:231–6.
3. Bafna PA, Balaraman R. Antiulcer and antioxidant activity of Pepticare, a herbomineral formulation. *Phytomedicine.* 2005;12:264–70. [[PubMed](#)]
4. Karnick CR. Pharmacology of Ayurvedic medicinal plants. Delhi: Sri Satguru Publications; 1996. pp. 1–3.
5. Vaidya BG. Nighantu Adarsa. 2nd ed. Varanasi: Chowkhamba Bharati Academy; 1998.
6. Jagetia GC, Baliga MS, Malagi KJ, Sethukumar Kamath M. The evaluation of the radioprotective effect of Triphala (an ayurvedic rejuvenating drug) in the mice exposed to γ -radiation. *Phytomedicine.* 2002;9:99–108. [[PubMed](#)]
7. Tripathi I. Amlapitta Chikitsa. In: Dwivedi R, Deo S, editors. Chakradatta of Shri Chakrapanidatta. 3rd ed. Varanasi: Chowkhamba Sanskrit Sansthan; 1997. pp. 45–54.
8. Naik GH, Priyadarsini KI, Bhagirathi RG, Mishra B, Mishra KP, Banavalikar MM, et al. *In vitro* antioxidant studies and free radical reactions of Triphala, an Ayurvedic formulation and its constituents. *Phytother Res.* 2005;19:582–6. [[PubMed](#)]
9. Kaur S, Michael H, Arora S, Harkonen PL, Kumar S. The *in vitro* cytotoxic and apoptotic activity of Triphala- an Indian herbal drug. *J Ethnopharmacol.* 2005;97:15–20. [[PubMed](#)]
10. Sabu MC, Kuttan R. Antidiabetic activity of medicinal plants and its relationship with their antioxidant property. *J Ethnopharmacol.* 2002;81:155–60. [[PubMed](#)]
11. Tripathi B. In: Sarngadhara Sanhita of Sri Sarngadharacharya. 4th ed. Varanasi: Chowkhamba Surbharati Prakashan; 2008. Kwathadi kalpana and Churna kalpana; p. 133. and 174.
12. The Ayurvedic Formulary of India, Part-1. 2nd ed. New Delhi: Department of ISM, Ministry of Health and Family Welfare, Government of India; 2003. Anonymous; pp. 103–10.
13. Ecobichon DJ. The basis of Toxicology testing. New York: CRC Press; 1997. pp. 43–86.
14. Parmar NS, Jagruti KD. A review of the current methodology for the gastric and duodenal antiulcer agents. *Indian J Pharmacol.* 1993;25:120–35.

15. Bafna PA, Balaraman R. Antiulcer and antioxidant activity of Normacid, a herbomineral formulation. Indian J Exp Biol. 2004;42:674–80. [\[PubMed\]](#)
16. Lowry OH, Rosebrough NJ, Farr AL, Randall RJ. Protein measurement with the Folin phenol reagent. J Biol Chem. 1951;193:265–75. [\[PubMed\]](#)
17. Ohkawa H, Ohishi N, Yagi K. Assay for lipid peroxides in animal tissues by thiobarbituric acid reaction. Anal Biochem. 1979;95:351–8. [\[PubMed\]](#)
18. Ohkawa H, Ohishi N, Yagi K. Reaction of linoleic acid hydroperoxide with thiobarbituric acid. J Lipid Res. 1978;19:1053–7. [\[PubMed\]](#)
19. McCord JM, Fridovich I. Superoxide dismutase: An enzymatic function of erythrocuprein.(hemocuprein) J Biol Chem. 1969;244:6049–55. [\[PubMed\]](#)
20. Sinha AK. Colorimetric assay of catalase. Anal Biochem. 1972;47:389–94. [\[PubMed\]](#)
21. Grunert RR, Phillips PH. A modification of nitroprusside method of analysis for glutathione. Arch Biochem. 1951;30:217–25. [\[PubMed\]](#)

Triphala Churna Powder is a really effective herbs in so many health conditions.

It one of the commonly used ayurvedic herbs.

If you are looking for a multipurpose ayurvedic health supplement then Triphala is best for you.

Triphala has many benefits and used in various ways for many different health condition. I ll guide you through it in this article.

8.What Is Triphala Churna Powder?

You must be wondering what is it.As the name suggest it is a natural blend of 3 very effective and beneficial herbs with numerous benefits.

This blend was first prescribed and used in “India”. It is made by mixing basically the extracts of Chebulic Myrobalan, Belleric Myrobalan and Indian Gooseberry.

This balanced formula was given or made some thousands of years back.

[Triphala](#) is also termed as ‘**nectar of life**’. In olden days it had some other names like Charaka Samhita and Susruta Samhita.

The Main Ingredients Of Triphala:

Triphala comprises of the three herbal fruits namely:

Indian Gooseberry:

Indian people know this fruit as AMLA. The scientific name of this fruit is Emblica Officinalis.

This is the main and the first ingredient of this herbal mixture. It is responsible for the detoxification effect of this mixture.

There are lots of benefits of this fruit. This fruit is rich in vitamin C which helps in boosting up the immunity level and is the best antioxidant.

Apart from this also works as a detoxifier by improving the digestion of food in the stomach.

As this fruit is rich in vitamin C thus it increases the absorption of the iron into the blood which is responsible for increasing the hemoglobin levels in the body.

It strengthens the reproductive system thus helps in regulating the menstrual cycles.

It also purifies the urinary system.it improves the brain functioning and also helps in supporting the cardiovascular system.

It increases the absorption of mineral due to which you get healthy hair and glowing face.It also strengthens the eye muscles and is very effective and beneficial for eyes.

Haritaki (Harad):

The other ingredient of this mixture is haritaki commonly called as harad. Its scientific name is Terminalia chebula .there are lots of benefits of this herb just like amla.

The best thing is this herb helps in preventing the eye diseases like cataract and glaucoma by strengthening the eye muscles.

This herb helps to cure the wound faster or heal it quickly. This herb helps in boosting up the immune system thus improving the memory.

This herb also helps you stay younger as it has anti aging properties. This is also very helpful for curing any kind of skin disease.

Bibhitaki(Baheda):

The third ingredient is the Bibhitaki which is commonly called as“Baheda”. Its scientific name is Terminalia Bellirica. This herb is well known for its blood purifying property.

It detoxifies the body and thus in this process removes the excess fat from the body too. This herb strengthens the hair root and is very effective in preventing hair graying.

It also improves the immunity of the body against viral and bacterial infections.

Triphala Churna Powder

How It Is Prepared?

Dried Haritaki, Bibhitaki, Indian Gooseberry are taken at equal quantity. The powdered form of this 3 ingredients are blended. This powder is commonly called as Triphala churna.

People in India recognize and take it for various health problems. Some use it for reducing weight and some to cure any problem related to the digestive system.

This powder(churna) comprising of the combination of the above herbs actually stimulates the metabolism thus it takes care of your stomach and digestive system.

Triphala Churna Benefits

Triphala **balances** and rejuvenates the three elements of body. **Vata** which regulates the nervous system and boosts the immunity.

Pitta which governs the metabolic activities and controls indigestion. **Kapha** which supports the structural integrity.

This herb is known to be the solution to solve many health conditions. Triphala churna is often referred as the complete health supplement.

Triphala has antioxidant, anti-inflammatory, antibacterial and antiviral, anti-cancer effects.

Due to its medicinal effects it can also cure anemia, jaundice, constipation, cough, asthma, fever, chronic ulcers, leucorrhoea, and pyorrhea.

Here is a list of the health benefits accompanied with this well-known “nectar of life” the Triphala churna.

Source: <http://mavcure.com/triphala-churna-benefits-ingredients-side-effects-dose-how-to-take/>